

PUBLISHED: 10-JUN-2016
2015.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - TDV8 4.4L DIESEL

COOLING SYSTEM PARTIAL DRAINING AND VACUUM FILLING (G1798818)

GENERAL PROCEDURES

26.10.08	COOLANT - PARTIAL DRAINING, FILLING AND BLEEDING	4400 CC, TDV8	0.8	USED WITHINS	
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SPECIAL TOOL(S)



HU-919

Coolant System Vacuum
Refill Kit

DRAINING



WARNING:

Since injury such as scalding could be caused by escaping steam or coolant, do not remove the filler cap from the coolant expansion tank while the system is hot.



CAUTIONS:

- The engine cooling system must be maintained with the correct concentration and type of anti-freeze solution to prevent corrosion and frost damage. Failure to follow this instruction may result in damage to the engine.
- Engine coolant will damage the paint finished surfaces. If spilt, immediately remove the coolant and clean the area with water.

**NOTE:**

Some variation in the illustrations may occur, but the essential information is always correct.

1.

**WARNING:**

Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

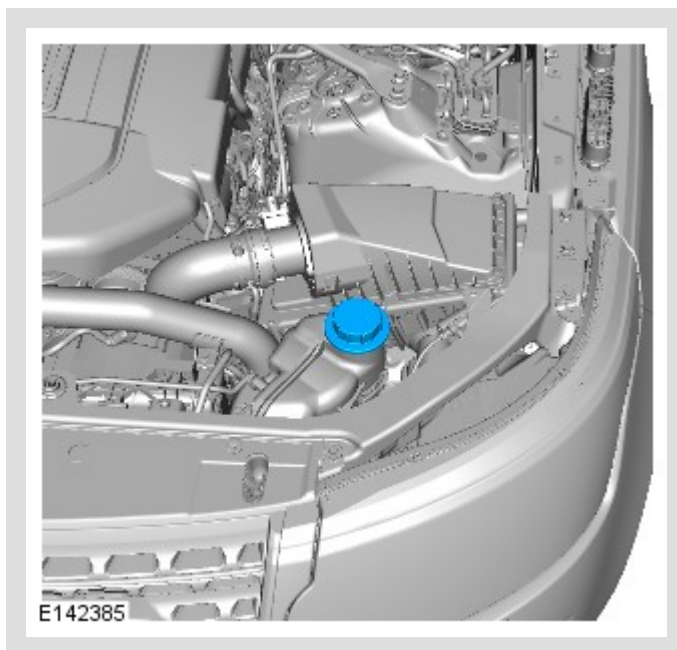
2.

**WARNING:**

When releasing the cooling system pressure, cover the coolant expansion tank cap with a thick cloth.

**CAUTION:**

Since injury such as scalding could be caused by escaping steam or coolant, make sure the vehicle cooling system is cool prior to carrying out this procedure.



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3. Remove engine undershield

Refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).

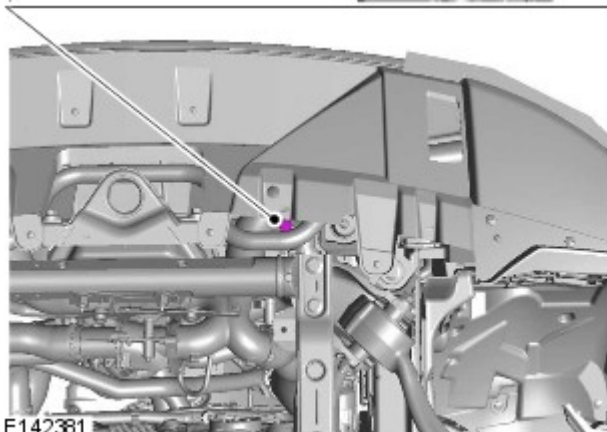
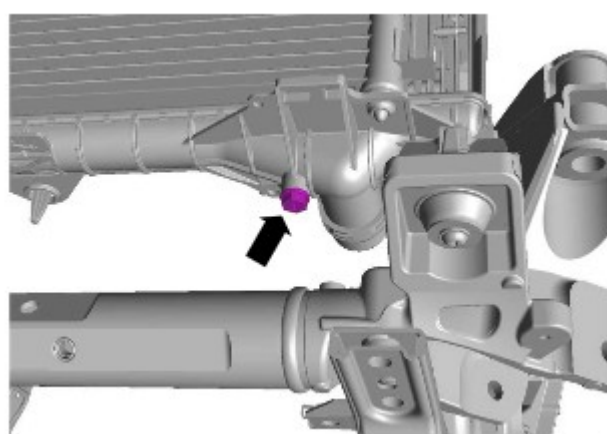
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- 4.

 CAUTION:

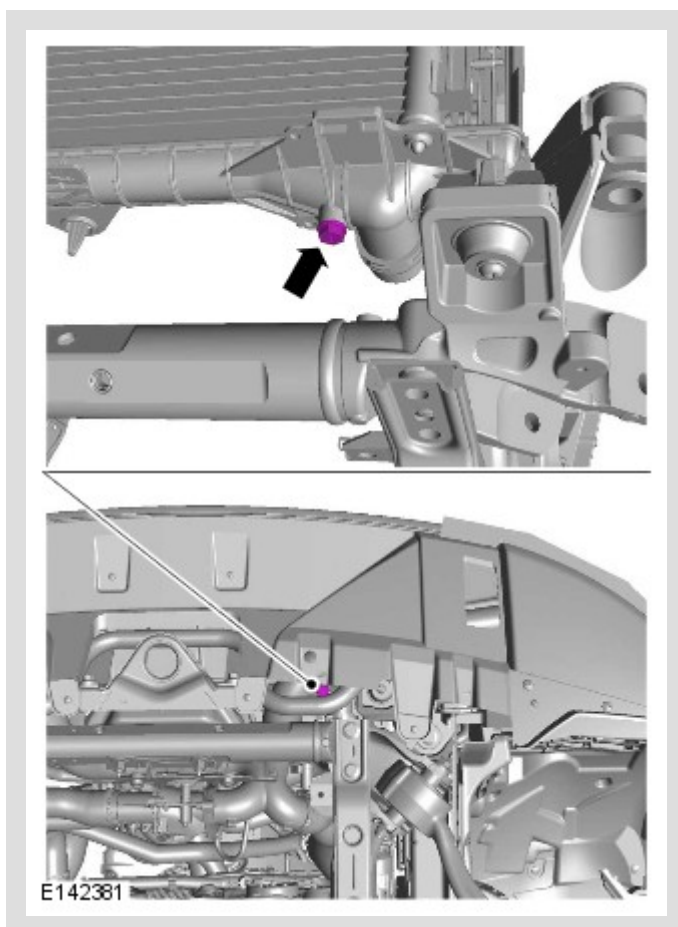
Be prepared to collect escaping coolant.

 NOTE:

Collect the coolant in a clean container and reuse.



E142381



Torque: 5 Nm

FILLING

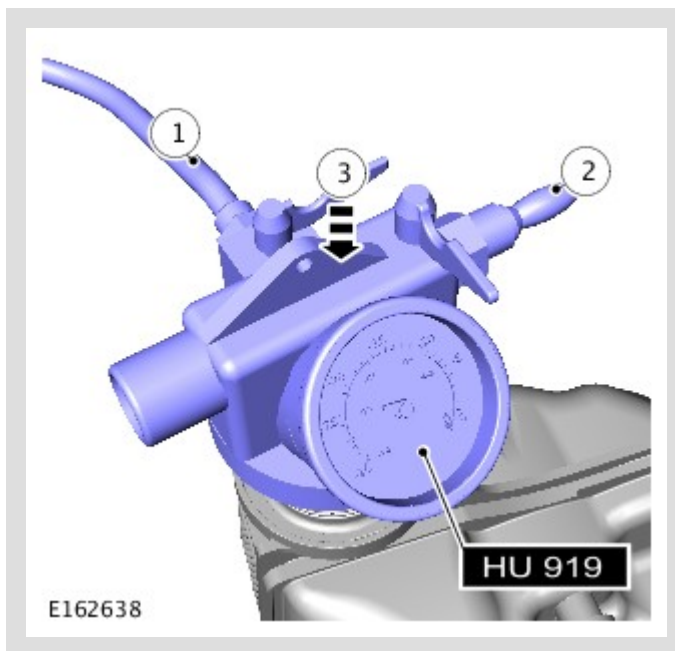
1. Install the engine undershield.
Refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).
2. Prepare a sufficient amount of coolant to the specified concentration.

3.



NOTES:

- Make sure the coolant supply valve is in the closed position on the special tool.
- The special tool needs an air pressure of 6 to 8 bar (87 to 116 psi) to operate correctly.
- Small diameter or long airlines may restrict airflow to the coolant vacuum fill tool.



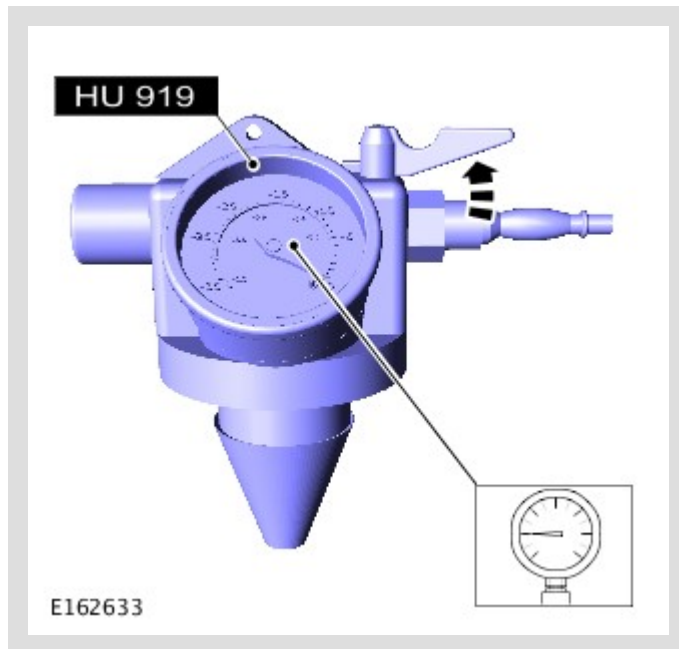
- *Special Tool(s):* [HU-919](#)
- Connect the hose from the vacuum refill adaptor into a container of clean coolant.
- Connect a regulated compressed air supply to the venturi tube assembly.
- Install the cooling system vacuum refill adaptor to the expansion tank.

4.



NOTES:

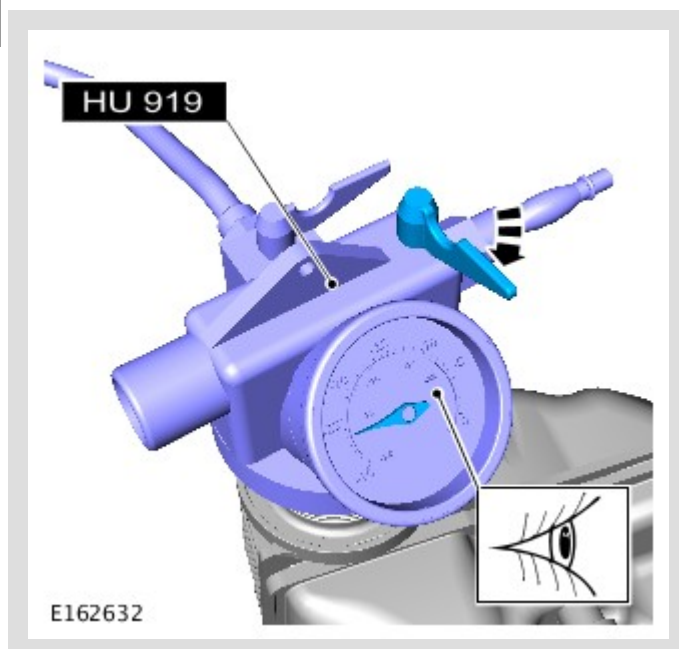
- Make sure the coolant supply valve is in the closed position on the vacuum filler gauge assembly.
- The coolant vacuum fill tool needs an air pressure of 6 to 8 bar (87 to 116 psi) to operate correctly.
- Small diameter or long airlines may restrict airflow to the coolant vacuum fill tool.



Open the air supply valve until -0.8 Bar is shown on the gauge.

Special Tool(s): [HU-919](#)

5.



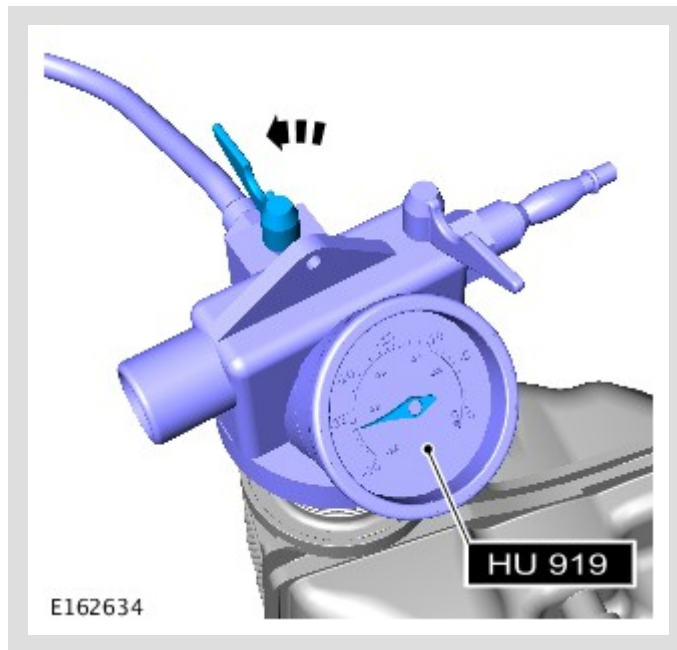
- *Special Tool(s):* [HU-919](#)
- Close the air supply valve.
- Allow 1 minute to check the vacuum is held.

6.



NOTES:

- The coolant is to be reused.
- Close the coolant supply valve when the coolant expansion tank MAX mark is reached or coolant movement has stopped.



Open the coolant supply valve and allow the coolant to be drawn into the system.

Special Tool(s): [HU-919](#)

7.

7.Remove the special tool.

8.

Connect exhaust extraction hoses to the tail pipes.

9.

Start and run the engine.

10.

Install the coolant expansion tank cap.

11.

Hold the engine speed at 2000 rpm until warm air is expelled from the heater.

12. Switch the engine off and allow to cool.

13. Clean any spilt coolant from the vehicle.

14.



WARNING:

Since injury such as scalding could be caused by escaping steam or coolant, allow the vehicle cooling system to cool prior to carrying out this procedure.

Check and top-up the coolant if required.